

NOTES FOR MATH 3150, WEEK 5

1. SUBSEQUENCES

Definition 1. Let $(x_n)_n$ be a sequence of real numbers. A subsequence of $(x_n)_n$ is a sequence of the form $(x_{n_k})_k$, where $n_1 < n_2 < n_3 < \dots$; that is, $(n_k)_k$ is strictly increasing in k .

Example.

Let $x_n = (-1)^n$. Then $(y_k)_k$ defined by $y_k = 1$ for all k is a subsequence of $(x_n)_n$ because $y_n = x_{2n}$.

We have the following result.

Theorem 2. Let $(x_n)_n$ be a sequence. Then

$$\lim_{n \rightarrow \infty} x_n = L$$

if and only if

$$\lim_{k \rightarrow \infty} x_{n_k} = L$$

for any subsequence $(x_{n_k})_k$ of $(x_n)_n$.

Proof. We prove the result in the case $L \in \mathbb{R}$, the cases $L = \infty$ or $L = -\infty$ are done similarly.

Assume first that

$$\lim_{n \rightarrow \infty} x_n = L.$$

Then for any $\varepsilon > 0$ there exists $N > 0$ so that

$$|x_n - L| < \varepsilon,$$

for any $n > N$.

Since $n_k \rightarrow \infty$ as $k \rightarrow \infty$, there exists $K > 0$ so that $n_k > N$, for all $k > K$. Therefore, in particular we get

$$|x_{n_k} - L| < \varepsilon,$$

for all $k > K$. This proves

$$\lim_{k \rightarrow \infty} x_{n_k} = L.$$

We now establish the converse, that if

$$\lim_{k \rightarrow \infty} x_{n_k} = L$$

for any subsequence $(x_{n_k})_k$ of $(x_n)_n$, then

$$\lim_{n \rightarrow \infty} x_n = L.$$

Assume by contradiction that this is not the case. Then there exists $\varepsilon > 0$ with the property that for any $N > 0$, there exists $n > N$ (depending on N) so that

$$(1) \quad |x_n - L| > \varepsilon.$$

We construct a subsequence $(x_{n_k})_k$ of $(x_n)_n$ as follows. We apply (1) for $N = 1$ and obtain $n_1 > 0$ so that

$$|x_{n_1} - L| > \varepsilon.$$

Now apply (1) for $N = n_1$ and obtain $n_2 > n_1$ so that

$$|x_{n_2} - L| > \varepsilon.$$

Continuing, we apply (1) for $N = n_k$ and obtain $n_{k+1} > n_k$ so that

$$|x_{n_{k+1}} - L| > \varepsilon.$$

By construction, the subsequence $(x_{n_k})_k$ does not converge to L , which is a contradiction. Therefore,

$$\lim_{n \rightarrow \infty} x_n = L.$$

□

We now present a fundamental result.

Theorem 3. *Any bounded sequence has a monotone subsequence.*

Proof. We say that $k \in \mathbb{N}$ is a peak point of $(x_n)_n$ if

$$x_k \geq x_m \quad \text{for all } m \geq k.$$

We have the following cases.

Case 1. $(x_n)_n$ admits an infinite number of peak points.

Denote them with $n_1 < n_2 < n_3 \dots$

Then the subsequence $(x_{n_k})_k$ is decreasing by construction. Indeed, n_1 being peak and $n_2 > n_1$, it follows that $x_{n_1} \geq x_{n_2}$. Continuing, we know that n_2 is peak and $n_3 > n_2$, therefore $x_{n_2} \geq x_{n_3}$ and so on.

Case 2. $(x_n)_n$ admits only finitely many peak points.

Then we can find $k_1 \in \mathbb{N}$ so that any $k \geq k_1$ is not a peak point of $(x_n)_n$.

By definition, there must exist $k_2 > k_1$ so that $x_{k_2} > x_{k_1}$.

But k_2 is not a peak point of $(x_n)_n$ either, so there must exist $k_3 > k_2$ so that $x_{k_3} > x_{k_2}$.

Continuing by induction, this produces an increasing subsequence of $(x_n)_n$.

By Case 1 and Case 2, this proves the theorem. □

As a consequence, we obtain the famous Bolzano-Weierstrass theorem.

Theorem 4. *Any bounded sequence has a convergent subsequence.*

Proof. The previous theorem states that any bounded sequence $(x_n)_n$ admits a monotone subsequence $(x_{n_k})_k$. The subsequence is also bounded, because $(x_n)_n$ is bounded. The monotone convergence theorem implies that $(x_{n_k})_k$ is convergent. $\square$

Definition 5. *A number L is called limit point for $(x_n)_n$ if there exists a subsequence $(x_{n_k})_k$ of $(x_n)_n$ so that*

$$L = \lim_{k \rightarrow \infty} x_{n_k}$$

The Bolzano-Weierstrass theorem implies that any bounded sequence has at least one limit point. However, if a sequence $(x_n)_n$ is not bounded above, then there exists a subsequence $(x_{n_k})_k$ that has limit $L = \infty$. Indeed, given $k \in \mathbb{N}$, assume we have constructed $x_{n_1}, \dots, x_{n_{k-1}}$. Then, as the sequence is unbounded above, there exists $n_k > n_{k-1}$ so that

$$x_{n_k} > k$$

By construction, we have that

$$\lim_{k \rightarrow \infty} x_{n_k} = \infty.$$

Similar considerations hold for sequences that are unbounded below, thus proving that the set of limit points of a sequence is not empty.

Example. The sequence $x_n = \frac{(-1)^n n}{n+1}$ has a subsequence $x_{2k} = \frac{2k}{2k+1}$, which converges to $L_1 = 1$. Another subsequence is $x_{2k+1} = -\frac{2k+1}{2k+2}$, which converges to $L_2 = -1$. Since $L_1 \neq L_2$, the sequence $(x_n)_n$ does not have a limit. The limit points of $(x_n)_n$ are $\{-1, 1\}$. Indeed, any converging subsequence of $(x_n)_n$ must have either finitely many even terms, or finitely many odd terms. In the former case, the limit of the subsequence must be $L_1 = 1$, in the latter case it is $L_2 = -1$.

2. CAUCHY SEQUENCES

Definition 6. *A sequence $(x_n)_n$ of real numbers is called Cauchy if for any $\varepsilon > 0$ there exists N so that*

$$|x_n - x_m| < \varepsilon$$

for all $n, m > N$.

Here are some immediate facts about Cauchy sequences.

Proposition 7. *We have the following properties:*

- (1) *Convergent sequences are Cauchy.*
- (2) *Cauchy sequences are bounded.*

Proof. **Proof of (1).**

Indeed, if $(x_n)_n$ is convergent, then any $\varepsilon > 0$ there exists N so that

$$|x_n - L| < \varepsilon$$

for all $n > N$.

Therefore, we also have

$$|x_m - L| < \varepsilon$$

for all $m > N$.

Adding up and using the triangle inequality we have

$$\begin{aligned} |x_n - x_m| &\leq |x_n - L| + |x_m - L| \\ &\leq 2\varepsilon \end{aligned}$$

for all $n, m > N$. This proves the result.

Proof of (2).

Since $(x_n)_n$ is Cauchy, then there exists N so that

$$|x_n - x_m| < 1$$

for all $n, m > N$.

In particular, this yields

$$\begin{aligned} |x_n| &\leq |x_n - x_N| + |x_N| \\ &\leq 1 + |x_N| \end{aligned}$$

for all $n > N$. Therefore, $(x_n)_n$ is bounded. $\square$

We use the above properties to establish a fundamental result about Cauchy sequences.

Theorem 8. *Any Cauchy sequence is convergent.*

Proof. If $(x_n)_n$ is Cauchy sequence, then it is bounded, and hence it admits a convergent subsequence by Bolzano-Weierstrass.

Let $(x_{n_k})_k$ be a subsequence convergent to L . We prove that $(x_n)_n$ converges to L as well.

Indeed, since $(x_n)_n$ is Cauchy, for any $\varepsilon > 0$ there exists $N > 0$ so that

$$|x_n - x_m| < \varepsilon$$

for all $n, m > N$.

As $(x_{n_k})_k$ converges to L , for any $\varepsilon > 0$ there exists $K > 0$ so that

$$|x_{n_k} - L| < \varepsilon$$

for all $k > K$.

Fix $k > K$ large enough so that $n_k > N$. Then by above we have

$$|x_n - x_{n_k}| < \varepsilon$$

for all $n > N$.

Applying the triangle inequality we conclude that

$$\begin{aligned} |x_n - L| &\leq |x_n - x_{n_k}| + |x_{n_k} - L| \\ &< 2\varepsilon \end{aligned}$$

for all $n > N$. This proves that $(x_n)_n$ converges to L . $\square$

Application. *We learn in calculus that if a series is absolutely convergent, then it is convergent as well. We explain this fact using Cauchy sequences.*

Indeed, assume that

$$\sum_{k=1}^{\infty} |a_k| < \infty.$$

This means that the sequence of partial sums

$$S_n = \sum_{k=1}^n |a_k|$$

is convergent. Being convergent, it is also Cauchy. Hence, for any $\varepsilon > 0$, there exists $N > 0$ so that

$$|S_n - S_m| < \varepsilon$$

for all $n, m > N$.

Without loss generality, we may assume that $n > m$. Then we have

$$S_n - S_m = \sum_{k=m+1}^n |a_k|.$$

By triangle inequality, the sequence of partial sums of $\sum_k a_k$, defined by

$$T_n = \sum_{k=1}^n a_k,$$

satisfies

$$\begin{aligned} |T_n - T_m| &= \left| \sum_{k=m+1}^n a_k \right| \\ &\leq \sum_{k=m+1}^n |a_k| \\ &< \varepsilon, \end{aligned}$$

for all $n > m > N$. This proves T_n is Cauchy, therefore T_n is convergent.

3. MORE EXAMPLES.

Problem. Define $(x_n)_n$ by $x_1 = 1$ and

$$x_{n+1} = 1 + \frac{1}{x_n}.$$

Prove that $(x_n)_n$ converges to $\frac{1+\sqrt{5}}{2}$. Hence,

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\dots}}} = \frac{1 + \sqrt{5}}{2}.$$

Proof. Checking a few values we get $x_1 = 1$, $x_2 = 2$, $x_3 = \frac{3}{2}$, $x_4 = \frac{5}{3}$.

Obviously, $(x_n)_n$ is not monotone. But we can prove by induction that the subsequence $(x_{2n+1})_n$ is increasing and the subsequence $(x_{2n})_n$ is decreasing.

First, let us prove by induction that $x_{2n-1} < \frac{1+\sqrt{5}}{2}$ and $x_{2n} > \frac{1+\sqrt{5}}{2}$, for all $n \geq 1$.

The case $n = 1$ follows directly from $x_1 = 1$ and $x_2 = 2$.

Assume that $x_{2n-1} < \frac{1+\sqrt{5}}{2}$ and $x_{2n} > \frac{1+\sqrt{5}}{2}$ and prove that $x_{2n+1} < \frac{1+\sqrt{5}}{2}$ and $x_{2n+2} > \frac{1+\sqrt{5}}{2}$.

Indeed, we have

$$\begin{aligned} x_{2n+1} &= 1 + \frac{1}{x_{2n}} \\ &< 1 + \frac{1}{\frac{1+\sqrt{5}}{2}} \\ &= \frac{1 + \sqrt{5}}{2} \end{aligned}$$

and then

$$\begin{aligned} x_{2n+2} &= 1 + \frac{1}{x_{2n+1}} \\ &> 1 + \frac{1}{\frac{1+\sqrt{5}}{2}} \\ &= \frac{1 + \sqrt{5}}{2}. \end{aligned}$$

This completes the induction step and shows that

$$x_{2n-1} < \frac{1 + \sqrt{5}}{2} < x_{2n}$$

for all $n \geq 1$.

Now we prove that $(x_{2n})_n$ is decreasing in n . We have

$$\begin{aligned} x_{2n+2} &= 1 + \frac{1}{x_{2n+1}} \\ &= 1 + \frac{1}{1 + \frac{1}{x_{2n}}} \\ &= \frac{2x_{2n} + 1}{x_{2n} + 1}. \end{aligned}$$

Therefore, $x_{2n+2} \leq x_{2n}$ if and only if

$$\frac{2x_{2n} + 1}{x_{2n} + 1} \leq x_{2n}.$$

Solving this, we get

$$x_{2n}^2 - x_{2n} - 1 \geq 0,$$

which is true because $x_{2n} \geq \frac{1+\sqrt{5}}{2}$.

This proves that $(x_{2n})_n$ is decreasing. Similarly, $(x_{2n-1})_n$ is increasing.

In conclusion, both have limits. Taking limit in the recurrence relation we get that $L = \frac{1+\sqrt{5}}{2}$. $\square$